

Unexpected shift from cyanobacterial to dinoflagellate dominance due to a summer drought

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ABSTRACT

The ruling paradigm is that future climate change scenarios will lead to an increase in the frequency, intensity, and duration of cyanobacterial blooms across the globe. It was therefore unexpected when, during an unusually warm and dry summer in southern Sweden in 2018, cyanobacteria did not dominate the phytoplankton community of the temperate Lake Vombsjön known for toxic cyanobacterial blooms. Instead, a potentially toxic dinoflagellate, *Naiadinium polonicum*, formed a large bloom and was dominant for the entire season. Dinoflagellates share important functional traits with cyanobacteria, but few studies have considered competition between dinoflagellates and cyanobacteria under future climate change scenarios. In order to understand the underlying factors leading to the *N. polonicum* bloom in 2018, we designed a study of Lake Vombsjön for an additional two years. When compared with subsequent years, 2018 was defined by an extended period of thermal stratification and hypoxia near the sediment surface. A significant positive relationship was also identified between *N. polonicum* biomass and both surface water temperature and soluble reactive phosphorus. We therefore suggest that the combination of higher temperatures and higher water column stability benefitted *N. polonicum* more so than cyanobacteria. Given that these factors are generally associated with cyanobacterial success, we conclude that freshwater phytoplankton community composition in temperate lakes can take different pathways in a warmer world.

1. Introduction

Lakes cover only a small portion of the Earth's continental land surface (approximately 3 %) (Downing et al., 2006) but provide extremely important ecosystem services such as drinking water, fishing, recreation and tourism. A major threat to these ecosystem services are harmful algal blooms (HABs) (Brooks et al., 2016). Though both pro- and eukaryotic organisms can cause HABs, the most common HABs in lakes are those dominated by cyanobacteria (Paerl, 1988). Problems associated with cyanobacterial harmful algal blooms (cyanoHABs) include the cells low edibility and nutritional value for zooplankton (Porter, 1977), increased turbidity and smothering of submerged vegetation (Scheffer et al., 1993), oxygen depletion caused by the microbial degradation of bloom biomass (Rabalais et al., 2010), and the production of secondary metabolites, some of which are toxic for aquatic organisms, pets, livestock and humans (Merel et al., 2013).

Climate change and extreme weather events are expected to be a potent catalyst for the expansion of cyanobacterial blooms (Carey et al.,

2012; Huisman et al., 2018; O'Neil et al., 2012). For temperate regions, climate change scenarios predict that lakes will experience increased temperatures, more intense and longer thermal stratification, as well as increased winter precipitation (Carey et al., 2012). The latter can increase the transportation of nutrients such as nitrogen and phosphorus, promoting cyanobacterial blooms in lakes (Paerl et al., 2016). Warmer temperatures have also been positively correlated with a higher proportion of cyanobacterial biomass in lakes (Kosten et al., 2012). This has led to the prediction that cyanobacteria will become more dominant with rising temperatures, in part because cyanobacteria tend to grow better at temperatures above 25°C than do eukaryotic species (Johnk et al., 2008; Robarts and Zohary, 1987). Warmer temperatures also allow for some species to expand their geographical range, thus altering species pools. For example, an increase in water temperature earlier in the growing season has been associated with the spread of the cyanobacterium *Raphidiopsis raciborskii* (basinonym *Cylindrospermopsis raciborskii*) from the tropics to temperate zones (Wiedner et al., 2007). Indirect effects of temperature, such as changes in thermal stratification

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and vertical mixing patterns, are at least as important as direct effects, since these processes govern the availability of both light and nutrients to phytoplankton (Diehl et al., 2002; Salmaso, 2005). Here too, cyanobacteria are predicted to have an advantage over other phytoplankton since they are able to control their buoyancy and move vertically through the water column, thus having access to light near the surface and nutrients from deeper layers (Visser et al., 1997).

One of the most pervasive bloom-forming cyanobacteria in freshwater ecosystems is *Microcystis*, which has been recorded in at least 108 countries worldwide (Harke et al., 2016). *Microcystis* can produce hepatotoxins called microcystins, which means that blooms are a potential risk to human health and ecosystem services particularly in lakes serving as drinking water supplies. For example, a *Microcystis* bloom that occurred in 2007 near the intake of a water plant in Lake Taihu, China, left approximately two million people without drinking water for at least a week due to high microcystin concentrations (Qin et al., 2010). Current predictions are that climate change will increase the occurrence and intensity of *Microcystis* blooms (Harke et al., 2016; Paerl and Huisman, 2009). Intense blooms of *Microcystis* have also been linked to extreme weather events such as droughts (Lehman et al., 2017). Due to the pervasiveness of *Microcystis* blooms and risks associated with them, there is an increasing need to understand factors causing *Microcystis* bloom formation in lakes and reservoirs, particularly with respect to climate change and extreme weather events.

The summer of 2018 was defined by a heatwave in Sweden leading to a hydrological drought (Teutschbein et al., 2022). During this summer, a sampling regime was initiated in Lake Vombsjön, which is a drinking water supply with a history of cyanobacterial blooms often dominated by *Microcystis* spp. in the summer, and *Planktothrix agardhii* in the fall (Alström et al., 2017; Johansson et al., 2019). The purpose of the sampling regime was to study patterns of *Microcystis* biomass and microcystin production in the lake, and link this to abiotic and biotic factors also monitored during sampling. Contrary to our expectations, *Microcystis* did not dominate the phytoplankton community in 2018. Instead a dinoflagellate, *Naiadinium polonicum* (synonyms: *Peridiniopsis polonicum*, *Peridinium polonicum* and *Glenodinium gymnodinium*) (Craveiro et al., 2015), was dominant during the entire summer. While *Naiadinium polonicum* had previously been identified in the lake in low numbers, no known blooms of this species had been reported. Early seasonal succession (Sommer et al., 1986) and functional trait models (Reynolds, 2006) have highlighted that cyanobacteria and dinoflagellates share a number of functional traits (e.g. large size, vertical motility) and therefore do well under similar conditions. This has also been observed in systems where cyanobacteria and dinoflagellates tend to co-occur (Grigorszky et al., 2019), or to outcompete one another (Crossetti et al., 2019; Reynolds and Bellinger, 1992; Vardi et al., 2002). In spite of this, few studies have considered how competition between cyanobacteria and dinoflagellates will affect cyanobacterial bloom dynamics under future climate change scenarios.

The shift in the summer phytoplankton community that occurred in Lake Vombsjön in 2018, from being cyanobacteria- to dinoflagellate-dominated, highlighted that current models predicting cyanobacteria to dominate in warmer weather are incomplete, especially for temperate regions. The aim of this study was therefore to address current gaps in the knowledge regarding shifts between cyanobacterial versus dinoflagellate dominance in temperate eutrophic lakes during high temperature events. To this end we identified the following questions: (i) How did the heatwave of 2018 affect pelagic abiotic conditions in Lake Vombsjön?; (ii) How did the extreme weather of 2018 affect planktonic community composition in Lake Vombsjön?; (iii) Did *Microcystis* and *N. polonicum* tend to be co-dominant or did one outcompete the other?; and (iv) Which abiotic factors best explained the success of *Microcystis* versus *Naiadinium polonicum*? To answer these questions, we designed a sampling program for an additional two years. We also utilized open-source meteorological and hydrological data. These data were then used to compare the drought year of 2018 with the two subsequent

years: 2019 and 2020. Existing historical data of the phytoplankton community was compiled to confirm the historical dominance of cyanobacteria in Lake Vombsjön.

2. Materials and method

2.1. Study sites, field survey, and historical data

Lake Vombsjön is a eutrophic lake located in southern Sweden (SWEREF 99: N55.68396, E13.58818). It has a surface area of 11.8 km² and its average and maximum depths are 6.6 m and 16 m respectively. The catchment area is 447 km² and is mainly agricultural. The lake is used both for drinking water and for recreation such as swimming and fishing.

Lake Vombsjön was sampled every two weeks in 2018 from late-June to mid-September. A more intensive sampling was then extended to 2019 and 2020, when samples were taken weekly from May to October. Sampling took place at a depth of ca 7 m (equivalent to average depth of the lake) at the same approximate location each year (SWEREF 99: N55.6775, E13.60889) (Fig. 1). Volume-integrated samples of the water column were taken using a 1-meter tube sampler (volume 1 L). In 2018 the top 2 m were sampled whereas in 2019 and 2020 the sampling depth was increased to 6 m to account for the fact that dinoflagellates can congregate in deeper layers (Lindstrom, 1992). From the integrated water sample, sub-samples were taken for determining nutrients, chlorophyll-a, pigment-based phytoplankton community composition and for quantitative phytoplankton samples. A quantitative zooplankton sample was collected by filtering 8-L of the integrated sample through a 45-µm plankton net. Quantitative phytoplankton and zooplankton samples were preserved with Lugol's solution for later enumeration. Surface temperature ($T_{1\text{ m}}$) was taken at 1 meter depth and Secchi depth was measured using a Secchi disk.

Samples were stored in a cooler and transferred to the lab. Unfiltered samples were divided into sub-samples for analyses of total phosphorus (TP) and total nitrogen (TN) and for determining phytoplankton community composition using spectral fluorometry. Samples for the analysis of soluble reactive phosphorus (SRP), nitrate plus nitrite ($\text{NO}_2 + \text{NO}_3$) and ammonium (NH_4) were filtered through GF/C glass microfiber filters (Whatman plc, Maidstone, United Kingdom). For chlorophyll-a, known volumes of water were filtered through GF/F glass microfiber filters (Whatman plc, Maidstone, United Kingdom). Filters were folded and wrapped in aluminum. All nutrient and chlorophyll-a samples were stored at -20°C until further analysis.

In addition to the field survey, historical phytoplankton community data, with a focus on historical cyanobacteria and dinoflagellate biomass in Lake Vombsjön, was compiled for years when sampling occurred including 1989–1995 from Cronberg (1996) as well as 1996–2002 and 2017 from Miljödata-MVM (2022). Reported data were based on quantitative phytoplankton samples collected in the month of August (summer bloom), except for 1991 for which there was no data for August and therefore data from September was reported. For the years 1989–2002 samples were taken from the top 2 m of the lake whereas in 2017 the top 8 m were sampled.

2.2. Sample analysis

A rough estimate of the overall phytoplankton community composition was measured directly after sampling using an AlgaeLabAnalyser (ALA) (bbe Moldaenke GmbH, Schwentinental, Germany). Briefly, percent chlorophyll originating from: green algae, cyanobacteria, diatoms plus dinoflagellates (the ALA cannot differentiate between diatoms and dinoflagellates) and cryptophytes, were determined based on pigment using light-emitting diodes with a visible range from ultraviolet to red. All nutrient samples were analyzed on a continuous segmented flow analyzer (AutoAnalyzer 3, SEAL Analytical GmbH, Norderstedt, Germany) at the Lake Erken analytical laboratory at

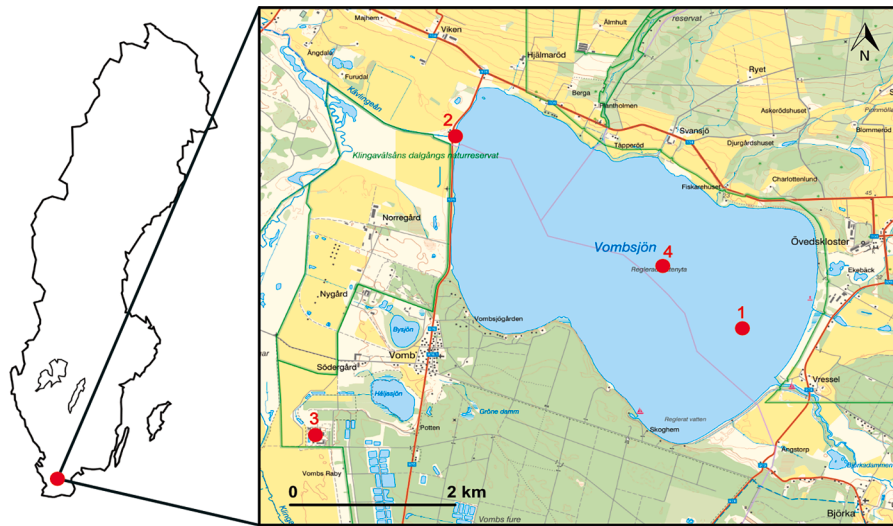


Fig. 1. Map of Lake Vombsjön located in southern Sweden. 1 corresponds to the sampling site, 2 corresponds to the site where outflow measurements were taken, 3 corresponds to the site where precipitation was measured, and 4 corresponds to deepest part of the lake. The map is modified from ©Lantmäteriet.

Uppsala University according to Swedish standards including TN (ISO 11,905–1), TP (ISO 6878:2005), SRP (ISO 6878:2005), $\text{NO}_3 + \text{NO}_2$ (ISO 13,395), and NH_4 (ISO 11,732:2005). Chlorophyll-*a* was measured according to Jespersen and Christoffersen (1987) on a laboratory fluorometer (TD-700, Turner Designs, Sunnyvale, California). Cyanobacteria and dinoflagellates were counted to the finest taxonomical level possible using an inverted microscope (Nikon Eclipse TS100, Tokyo, Japan) in 5 ml settling chambers according to the Utermöhl method (Edler and Elbrächter, 2010). Cells per colony of *Microcystis* species were counted using a grid reticle and a previously determined number of cells per grid as described in Ekvall et al. (2013). Biovolumes were determined according to Hillebrand et al. (1999) except for *Ceratium hirundinella* whose biovolume was estimated according to Napiorkowska-Krzebietke and Kobos (2016). Cell dimensions for biovolume calculations for the entire season were determined by measuring the first ten species on the date they were most abundant that year. For colonies, five cells from the first ten colonies (a total of fifty cells) were measured on the date they were most abundant that year. Assuming that cytoplasm density is equal to water (1.0 g mL^{-1}), biomass (wet weight, mg L^{-1}) was estimated assuming $1 \text{ mm}^3 \text{ L}^{-1} = 1 \text{ mg L}^{-1}$. Zooplankton were counted to the finest taxonomical unit possible, and length measured using an inverted microscope (Olympus CK30, Tokyo, Japan). Cladoceran and copepod (excluding nauplii) biomass (dry weight, mg L^{-1}) was calculated from length of the first 20 units counted according to EPA (2016). Rotifer biomass (dry weight, mg L^{-1}) was calculated from biovolumes reported in Aasa (1970) assuming cytoplasm density is equal to water, and that dry weight is 10 % of the wet weight (EPA, 2016). Zooplankton samples were counted every second week during the sampling season for all three years.

2.3. Data analysis

All nutrient concentrations above the laboratory detection limit, but below the quantification limit were given a value of half the quantification limit for further analysis. Dissolved inorganic nitrogen (DIN) was calculated by summing $\text{NO}_3 + \text{NO}_2 + \text{NH}_4$. TN:TP and DIN:SRP were calculated using molar ratios. The euphotic zone Z_{eu} (m) was calculated as 1.7 times the Secchi depth (Jensen et al., 1994). Additional information, including dissolved oxygen (DO, mg L^{-1}) and temperature profiles (May–October) for the deepest part of the lake (see Fig. 1), were provided by the local drinking water plant (Vombverket, Sydvatten AB) for all three years. Profiles started at 1 meter below the surface and ended within 1 meter of the sediments (exact location above sediments

were not defined). Temperature profiles were used to calculate relative water column stability (RWCS), which is a dimensionless parameter positively correlated with thermal stratification (Yang et al., 2016): $\text{RWCS} = (\rho_b - \rho_w) / (\rho_4 - \rho_5)$. In the formula, ρ_b and ρ_w represent the bottom water density and the water density at a certain depth (in this case $T_1 \text{ m}$), respectively, whereas ρ_4 and ρ_5 represent water density at 4°C and 5°C , respectively. Water density was calculated from the empirical formula below (Martin and McCutcheon, 1998):

$$\rho_T (\text{kg m}^{-3}) = 1000 \left[1 - \frac{(T + 288.9414)}{508929.2(T + 68.12963)} (T - 3.9863)^2 \right]$$

Precipitation (Prec, mm) and river discharge (Q_{out} , $\text{mm}^3 \text{ s}^{-1}$) data were downloaded from the Swedish Meteorological and Hydrological Institute (SMHI., 2022) for station number 53,410 (SWEREF 99: N55.6629, E13.5288) and station number 2018 (SWEREF 99: N55.6984, E13.5551), respectively (see Fig. 1). Total global radiation (G_{rad} , $\text{kJ m}^{-2} \text{ day}^{-1}$) data were downloaded from the Agri4Cast Data portal of the Joint Research Centre (JRC) which contains daily meteorological information from weather stations interpolated on a $25 \times 25 \text{ km}$ grid (Toreti, 2014).

Cyanobacteria species were grouped at the genus level with the exception of Oscillatoriales which was grouped at the order level. Dinoflagellate species not belonging to *Ceratium hirundinella* or *Naiadinium polonicum*, were grouped as ‘other dinoflagellates’ due to their, collectively, relatively low biomass. Zooplankton species were grouped into five categories based on their taxa: rotifers, small cladocerans (*Bosmina*, *Ceriodaphnia*, *Chydorus*), large cladocerans (*Diaphanosoma*, *Daphnia*), calanoid copepods and cyclopoid copepods.

May phytoplankton biomass was excluded from statistical analyses due to very low counts (or no counts) and therefore low precision. To analyze variation of cyanobacteria and dinoflagellate community composition, a Principal Component Analysis (PCA) was used from the R package ‘vegan’, version 2.5–7 (Oksanen et al., 2020; RStudio Team, 2020). Given that species biomass spanned three orders of magnitude, species biomass data were transformed using the equation $y' = \log_{10}(10y + 1)$, which is suitable when there are many small values (Lepš and Šmilauer, 2003). Environmental variables and zooplankton biomass were fitted onto two separate ordination diagrams using the function envfit with 999 permutations and were automatically standardized in vegan. The abiotic variables included in this analysis were $T_1 \text{ m}$, Z_{eu} , TN, TP, TN:TP, SRP, $\text{NO}_3 + \text{NO}_2$, NH_4 , Prec, Q_{out} and G_{rad} . Q_{out} and G_{rad} were based on 30-day averages prior to sampling, which had the highest overall correlation to either *Naiadinium polonicum* or *Microcystis* biomass

when compared to 7- and 15-day averages. Precipitation (Prec) was based on 30-day cumulative values prior to sampling for the same reason as above. RWCS and DO_{bottom} could not be included in the analysis since data collection of these variables did not correspond with the dates of our sampling. The ordination diagram with zooplankton biomass (rotifer, small cladocerans, large cladocerans, calanoid copepods and cyclopoid copepods) includes results from every second week when zooplankton were counted.

To compare biomass of *Naiadanium polonicum* and *Microcystis* over a season, area under the curve (AUC) was calculated for both species and then averaged over the sampling season. The calculation of AUC was performed in DataGraph 5.0.

To explore the relationship between *Naiadanium polonicum* and *Microcystis* biomass with environmental variables ($T_{1\text{m}}$, Z_{eu} , TN, TP, TN:TP, SRP, $\text{NO}_3 + \text{NO}_2$, NH_4 , Prec, Q_{out} and G_{rad}), a multiple linear regression was performed within the software R (RStudio Team, 2020), along with model averaging using the package MuMIn, version 1.43.17 (Barton, 2009). *Naiadanium polonicum* and *Microcystis* biomass were log transformed as described above using the equation $y' = \log_{10}(10y + 1)$. In order to have comparable coefficients, environmental variables were standardized according to the equation $Z = (x - \mu)/\sigma$, where Z is the z-score value, x is the value being standardized, μ is the mean and σ is the standard deviation. Collinearity among environmental variables were assessed using variance inflation factors (VIFs) from the initial model where all variables were included. Initial VIFs were high (>10) for multiple variables (TN, $\text{NO}_3 + \text{NO}_2$, TN:TP, TP, G_{rad}). Pearson correlation coefficients between variables were then assessed. TN, $\text{NO}_3 + \text{NO}_2$ and TN:TP were all highly correlated (coefficients > 0.8) with each other, TP was highly correlated with SRP and G_{rad} was highly correlated with $T_{1\text{m}}$. Since we were most interested in the effect of temperature and inorganic nutrients on *Naiadanium* and *Microcystis* biomass, TN, TP, TN:TP and G_{rad} were removed from the analysis. The final list of included environmental variables were therefore: $T_{1\text{m}}$, Z_{eu} , SRP, $\text{NO}_3 + \text{NO}_2$, NH_4 , Prec and Q_{out} . VIFs were subsequently ≤ 4 . Akaike Information Criterion (AIC) was used for model averaging. For both *Naiadanium polonicum* and *Microcystis* spp., models where delta AIC was <2 were included for determining averaged variable estimates (Barton, 2009). The models were checked for autocorrelation, homoscedasticity and normality of the residuals. Variable coefficients, standard error and p-values were reported.

3. Results

3.1. Physical and chemical parameters

Surface water temperatures ($T_{1\text{m}}$) during the three sampling seasons (May to October) ranged from 10.5°C to 24.3°C (Fig. 2a). Monthly average $T_{1\text{m}}$ was higher for the entire sampling season in 2018 compared with 2019 and 2020, though most notably in May and July (Table 1). In May, average $T_{1\text{m}}$ was about 3°C higher in 2018 (16.1°C) compared with 2019 (12.9°C) and 2020 (13.0°C). In July, average $T_{1\text{m}}$ was highly variable between years, with a value of 22.3°C in 2018, 20.2°C in 2019 and 18.6°C in 2020. The highest recorded temperature occurred in 2018 (24.3°C), and was 2.1°C and 1.3°C warmer than the highest recorded temperatures in 2019 (22.2°C) and 2020 (23°C) (Fig. 2a). Relative water column stability (RWCS) was also higher in 2018 compared with 2019 and 2020 for the period May to July (Fig. 2b) implying stronger thermal stratification and reduced vertical mixing. Dissolved oxygen levels measured near the sediment surface (DO_{bottom}) ranged from 0 to 11.75 mg L^{-1} (Fig. 2c). In contrast to 2019 and 2020, DO_{bottom} was below 2 mg L^{-1} for an extended period during the 2018 sampling season, indicating hypoxic conditions for all of June and July (Fig. 2c).

Cumulative precipitation was particularly low in May, July and September of 2018 when compared with 2019 and 2020 (Fig. 3a). For the period May to July 2018, total rainfall was 67.9 mm compared to 131.8 mm in 2019 and 125.2 mm in 2020, in other words approximately

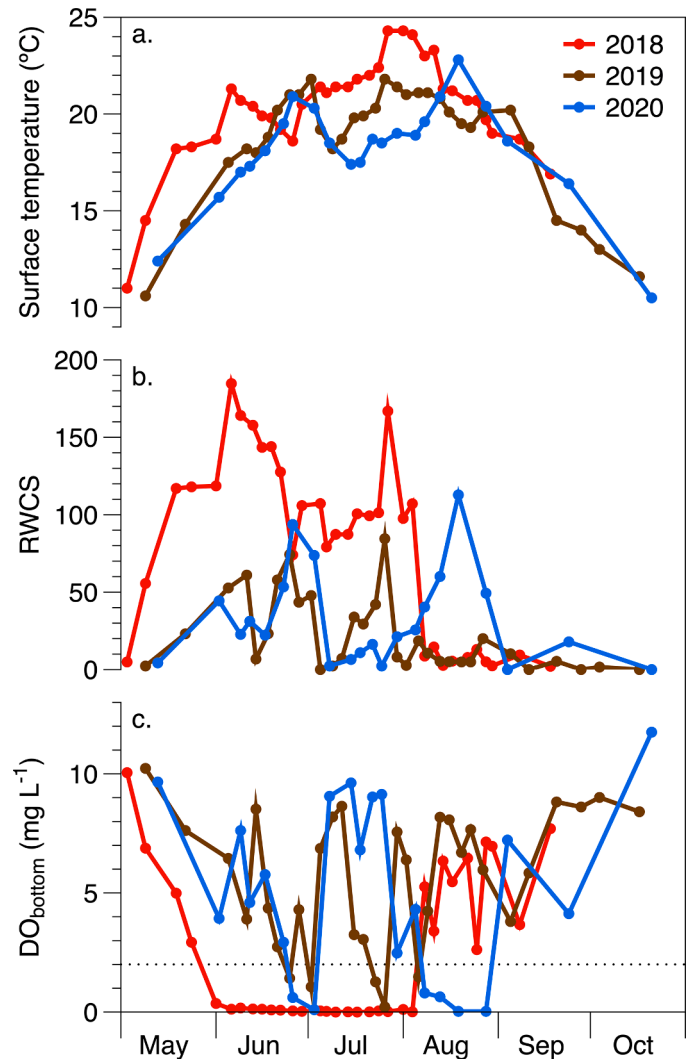


Fig. 2. Water treatment plant (Vomberket) data for (a) surface water temperature, (b) relative water column stability (RWCS) and (c) dissolved oxygen (DO_{bottom} , measured near the sediment surface). Measurements were taken May–October at the deepest part of the lake. The dotted line in (c) corresponds to a dissolved oxygen concentration of 2 mg L^{-1} . Conditions below this line are considered hypoxic.

Table 1

Monthly mean surface water temperatures (°C) and standard deviation from combined data (Vomberket drinking water treatment plant and lake sampling) for May–October in Lake Vombsjön.

	May	June	July	Aug	Sept	Oct
2018	16.1 ± 3.3	19.9 ± 1.1	22.3 ± 1.3	21.6 ± 1.6	17.8 ± 0.9	No data
2019	12.9 ± 1.9	19.3 ± 1.5	20.2 ± 1.3	20.5 ± 0.7	17.0 ± 3.1	12.3 ± 0.9
2020	13.0 ± 1.4	18.6 ± 1.7	18.6 ± 0.9	20.8 ± 1.3	17.4 ± 1.0	11.7 ± 1.8

half (Supporting Information Table S1). Cumulative precipitation in the spring (January – April) was, however, similar all three years (Supporting Information Table S1). Average outflow from Vombsjön was also lower in 2018. This was first observed in July and then continued for the rest of the summer (Fig. 3b).

In terms of nutrients, 2018 differed from the following years, in that both total nitrogen (TN) and total phosphorus (TP) concentrations were lower in 2018 compared with 2019 and 2020 (Fig. 4a, d). However, TN:

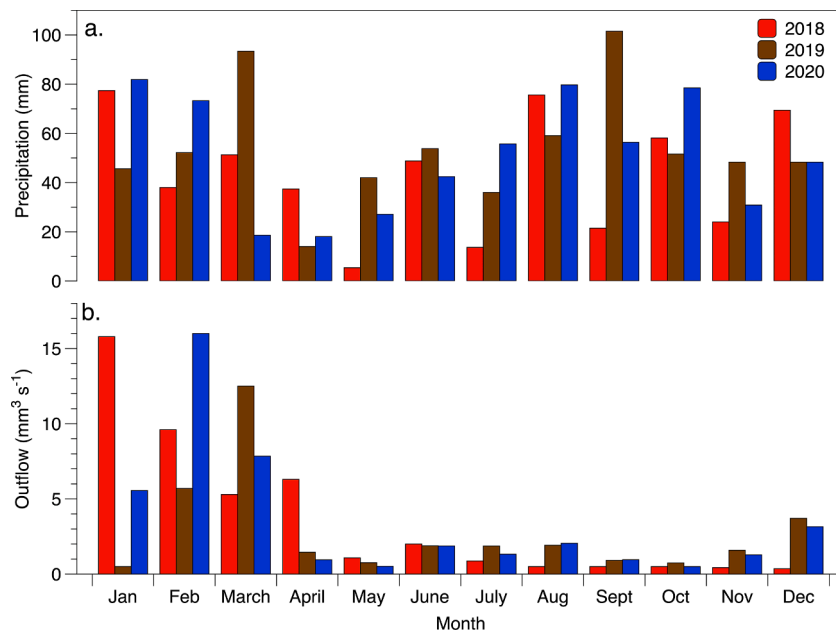


Fig. 3. Monthly cumulative precipitation and average outflow for Vombsjön 2018–2020 measured at station 53,410 (SWEREF 99: N55.6629, E13.5288) and station 2018 (SWEREF 99: N55.6984, E13.5551) respectively (see Fig. 1).

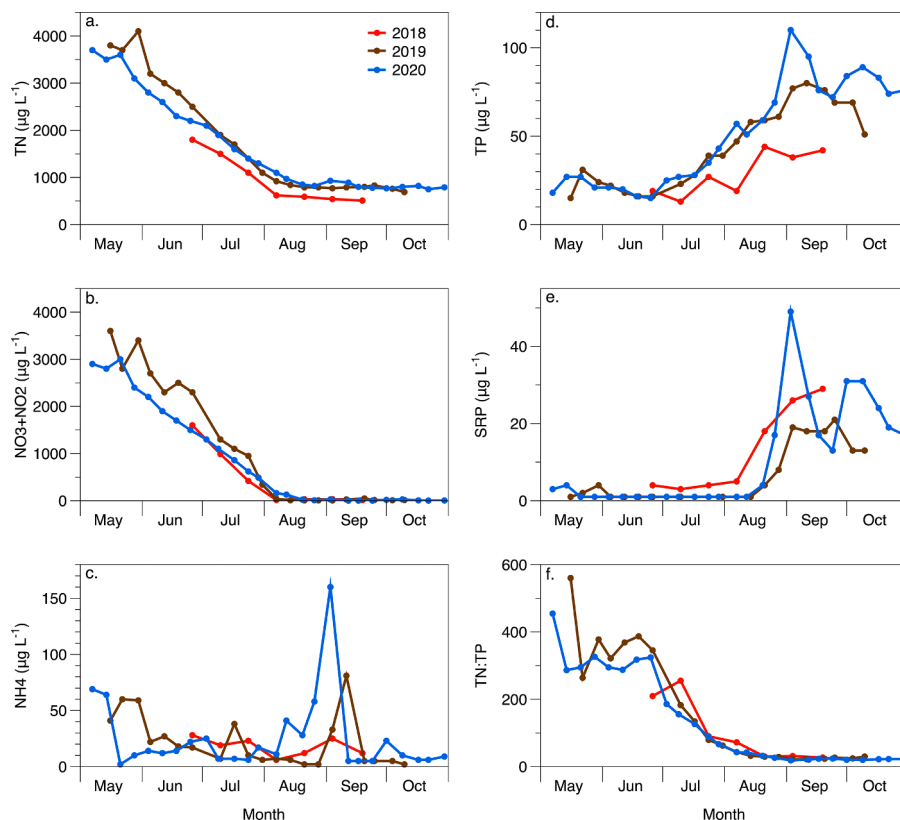


Fig. 4. Nutrient concentrations from lake sampling for 2018 (June–September), 2019 (May–October) and 2020 (May–October). TN = total nitrogen, NO₃+NO₂ = nitrate + nitrite, NH₄ = ammonium, TP = total phosphorus, SRP = soluble reactive phosphorus. Values under the detection limit were plotted as half the detection limit.

TP ratios were similar between years (Fig. 4f). Nitrate plus nitrite (NO₃+NO₂) concentrations were slightly higher in 2019 compared with 2018 and 2020 from May to late-July (Fig. 4b), while no obvious trends were observed in ammonium (NH₄) concentrations within or between years (Fig. 4c). Despite lower TP concentrations, soluble reactive

phosphorus (SRP) concentrations were higher in 2018 compared to 2019 and 2020 (Fig. 4e). There was, however, a notable peak in both SRP and NH₄ concentrations in early September 2020, which seemed to coincide with a mixing event that followed a 4-week period of hypoxia (Fig. 2c). In mid-August a switch was observed all years from what

appeared to be excess $\text{NO}_3 + \text{NO}_2$ concentrations to excess SRP concentrations suggesting a switch in the system from being P- to N-limited (Fig. 4b and 4e). This switch coincided with DIN:SRP concentrations below the Redfield ratio of 16 (Supporting Information Table S2).

3.2. Phytoplankton and zooplankton biomass and community composition

3.2.1. Historical phytoplankton community

According to historical data collected between 1989 and 2002, as well as 2017, cyanobacteria were dominant (> 50 % of the total phytoplankton community) during the month of August in 11 out of 15 years (Supporting Information Table S3). *Microcystis* spp. dominated the phytoplankton community 4 out of 6 years from 1989 to 1994, but then only once in the years following that are on record. Other cyanobacteria species that dominated included *Planktothrix agardhii* (1989, 1996 and 1997) and *Woronichinia naegliana* (2000 and 2017). In years where cyanobacteria were not dominant, the dominant taxa (taxa with the largest biomass) were the diatoms *Stephanodiscus* spp. and *Aulacoseira* spp. in 1992 and 1998 respectively. The dinoflagellate *Ceratium hirundinella* was dominant in 1999 (but didn't make up >50 % of the phytoplankton community) and the cryptomonad *Cryptomonas* spp. in 2001. *Naiadinium polonicum* was reported as early as 1999 but only ever with low biomass (<0.07 mg L^{-1}) (Supporting Information Table S3).

3.2.2. 2018–2020 sampling

The maximum chlorophyll-a value in 2018 (126 $\mu\text{g L}^{-1}$) was approximately 2–2.5 times higher than the maximum values in 2019 (50 $\mu\text{g L}^{-1}$) and 2020 (67 $\mu\text{g L}^{-1}$) (Fig. 5a). According to a pigment-based analysis of phytoplankton (ALA), during 2018 diatoms plus dinoflagellates made up >50 % of the community from early-August until the final sampling date on September 3rd (Fig. 5b). Based on microscopy we could establish that *N. polonicum* contributed most to the group diatoms plus dinoflagellates and that *N. polonicum* was dominant already from the end of July. In 2019, cyanobacteria were dominant from mid-July onward, only dipping below 50 % of the community on a couple of sampling occasions (Fig. 5c). From September 10th 2019 onwards, cyanobacteria made up approximately 80 % or more of the community. In 2020, there was no clearly dominant group (Fig. 5d). Green algae made up >50 % of the community on multiple sampling dates, but these events were highly fluctuating and spread out throughout the sampling

season. In 2020, cyanobacteria never made up >40 % of the community.

To better understand population dynamics between cyanobacteria and dinoflagellates, both groups were counted on a finer taxonomical level and are henceforth referred to as the 'counted community'. Identified taxa are presented in detail in Supporting Information Table S4. The most dominant taxa from the counted community, defined here as taxa whose biomass exceeded 1 mg L^{-1} during at least one of the growing seasons, were *Microcystis* spp., *Aphanizomenon* sp., *Dolichospermum* spp., Oscillatoriales, *Naiadinium polonicum* and *Ceratium hirundinella*, and are plotted in Fig. 6.

In 2018, *N. polonicum* was dominant from July 23rd onward, reaching a maximum biomass of 20.8 mg L^{-1} on August 20th (Fig. 6a). Maximum *C. hirundinella* biomass (1.8 mg L^{-1}) in 2018 was also more than twice that of 2019 (0.5 mg L^{-1}) or 2020 (0.2 mg L^{-1}) (Fig. 6a-c). In 2019, *C. hirundinella* dominated the counted community in June, reaching its maximum biomass on July 9th (0.5 mg L^{-1}) (Fig. 6b). After July 9th, *Microcystis* spp. took over as the dominant species. The peak biomass for *Microcystis* spp. in 2019 was on August 6th (1.6 mg L^{-1}). After August 6th, *Microcystis* spp. remained the dominant species in the counted community until mid-September, when *Dolichospermum* spp. took over. Maximum *Dolichospermum* spp. biomass in 2019 occurred on September 24th (1.1 mg L^{-1}) (Fig. 6b). In 2020, Oscillatoriales and *Aphanizomenon* sp. dominated the counted community on August 6th when their respective biomasses peaked at approximately 1 mg L^{-1} and then fell quickly (Fig. 6c). Following August 6th, *N. polonicum* and *Dolichospermum* spp. biomass increased in tandem. The maximum biomass for both *N. polonicum* and *Dolichospermum* spp. in 2020 was measured on September 23rd and was 1.8 mg L^{-1} (Fig. 6c).

Peak zooplankton biomass was 1.5 mg L^{-1} in 2018 compared with 0.6 mg L^{-1} in 2019 and 0.7 mg L^{-1} in 2020 (Fig. 6d-f). In 2018, the zooplankton community was dominated by copepods. Cyclopoid copepod biomass was larger in the beginning of the season whereas calanoid copepod biomass was larger towards the end of the season (Fig. 6d). In 2019, there was no clearly dominant group of zooplankton, though large cladocerans had the highest peak biomass for the season on September 3rd (0.3 mg L^{-1}) (Fig. 6e). In 2020, rotifers dominated from June to mid-August followed by small cladocerans from mid-August to September (Fig. 6f).

Principal component analyses (PCAs) were used to visualize variability in species composition based on the entire counted community. Identified cyanobacteria included *Aphanizomenon* sp., *Cuspidothrix* sp.,

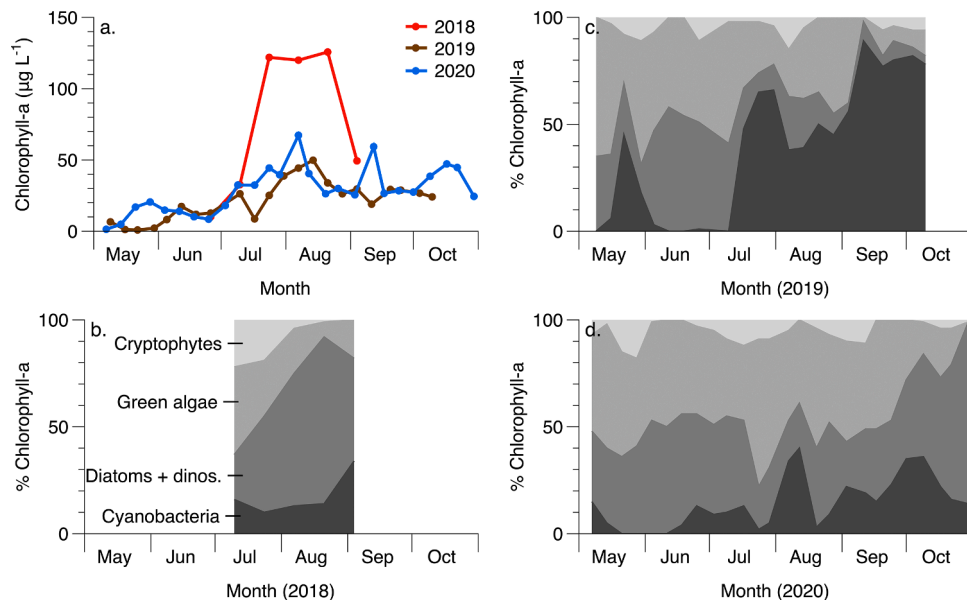


Fig. 5. (a) Chlorophyll-a (from fluorometry) and (b-d) phytoplankton community composition (from AlgaeLabAnalyser) for 2018, 2019 and 2020.

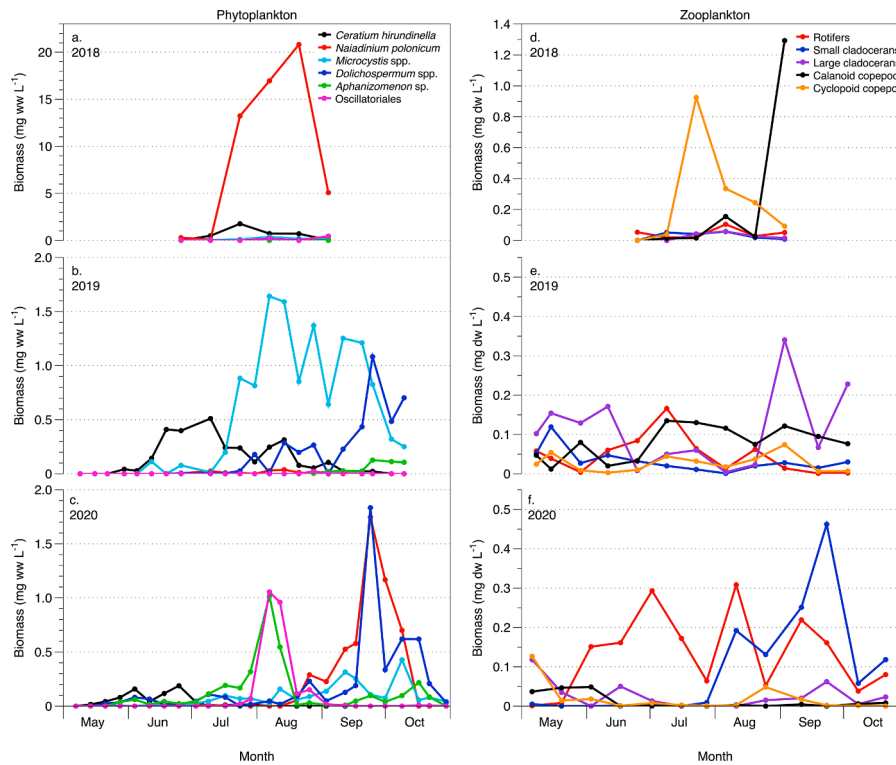


Fig. 6. (a-c) Phytoplankton (cyanobacteria and dinoflagellates) and (d-f) zooplankton biomass for 2018 (June–September), 2019 (May–October) and 2020 (May–October). Only dominant phytoplankton genera/species were plotted, ww = wet weight. Zooplankton were plotted by group, dw = dry weight. Y-axis scale for 2018 is different from 2019 to 2020 for both phytoplankton and zooplankton.

Dolichospermum spp., *Microcystis* spp., *Woronichinia* spp. and *Oscillatoriales*. Identified dinoflagellates included *Ceratium hirundinella*, *Naidinium polonicum* and ‘other dinoflagellates’. Environmental variables (Fig. 7a) and zooplankton biomass (Fig. 7b) were then fitted onto the ordination diagrams as vectors. Since zooplankton were only counted every second week, phytoplankton community data in Fig. 7b represents every second week of sampling and are thus fewer. In Fig. 7a and 7b, the first two axes of the PCA could explain 66.1 % and 71.8 % of the variability in community composition, respectively. There was a clear separation of community composition by year, in particular for 2018. *N. polonicum* was most prominent along the first axis (PC1) in both plots (Fig. 7a and b), whereas the remaining species were generally spread out along the second axis (PC2). *Microcystis* spp. appeared to have

a strong influence on PC2 in Fig. 7a but less so in Fig. 7b where *Dolichospermum* spp. and *C. hirundinella* had the strongest influence on PC2. Based on the angles between species and explanatory variables, *N. polonicum* was most positively correlated with the environmental variables surface water temperature (T_{1m}) and soluble reactive phosphorus (SRP), and most negatively correlated with outflow (Q_{out}) (Fig. 7a). None of the included environmental variables came through as having a particularly strong influence on PC1 (Fig. 7a), whereas total phosphorus (TP), nitrate plus nitrite (NO_3+NO_2), total nitrogen (TN) and TN:TP ratios all had a strong influence on PC2 (Fig. 7a). *Microcystis* was positively correlated with TP and negatively correlated with NO_3+NO_2 , TN and TN:TP (Fig. 7a). In Fig. 7b, PC1 was strongly influenced by cyclopoid copepods, with which *N. polonicum* was positively

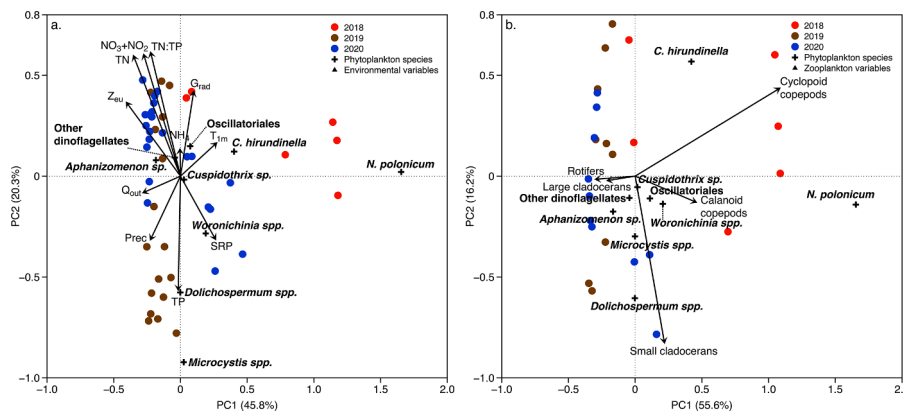


Fig. 7. Principal component analysis of the phytoplankton community composition (cyanobacteria and dinoflagellates in bold font). (a) All phytoplankton data, environmental factors were fitted onto the ordination graph using the function envfit. (b) Data from every second week when zooplankton biomass were counted, zooplankton biomass were fitted onto the ordination graph using the function envfit. In both graphs, length of arrows indicates strength of predictors. T_{1m} = surface water temperature, Z_{eu} = euphotic zone, TN = total nitrogen, TP = total phosphorus, SRP = soluble reactive phosphorus, NO_3+NO_2 = nitrate + nitrite, NH_4 = ammonium, TN:TP = total nitrogen to total phosphorus ratio, Prec = 30-day cumulative precipitation, Q_{out} = 30-day average outflow and G_{rad} = 30-day average global radiation..

correlated. PC2 was strongly influenced by small cladocerans with which *Microcystis* (and *Dolichospermum* spp.) was positively correlated (Fig. 7b).

3.3. Dynamics of *Naiadinitium polonicum* and *Microcystis* spp. 2018–2020

To evaluate patterns of abundance of *Microcystis* spp. and *N. polonicum* over a complete sampling season, area under the curve (AUC) was calculated for biomass and then averaged by the number of days in the sampling season. In 2018, averaged AUC for *N. polonicum* was approximately 70x larger than that of *Microcystis* spp.: 10.7 mg L^{-1} compared with 0.15 mg L^{-1} respectively (Fig. 8a). Likewise, averaged AUC for *Microcystis* spp. in 2019 was approximately 70x larger than that of *N. polonicum*: 0.67 mg L^{-1} compared with 0.01 mg L^{-1} respectively (Fig. 8b). Average AUC for both species in 2020 were lower compared to the previous years: 0.26 mg L^{-1} for *N. polonicum* and 0.10 mg L^{-1} for *Microcystis* spp. (Fig. 8c).

A multiple linear regression with model averaging was used to determine which environmental variables had the largest effect on *N. polonicum* and *Microcystis* spp. biomass. The results showed that there was a statistically significant relationship between *N. polonicum* and the environmental variables SRP, T_{1m} and Q_{out} (Table 2). *N. polonicum* biomass was positively correlated with SRP and T_{1m} (coefficients of 0.49 and 0.48 respectively) and negatively correlated with Q_{out} (coefficient of -0.31). *Microcystis* spp. was negatively correlated with NO_3+NO_2 (coefficient of -0.23) but had no statistically significant relationship with any other variable included in the analysis.

4. Discussion

The ruling paradigm is that future climate change scenarios including increased temperatures, more variable precipitation and nutrient loading, as well as more intense and longer periods of thermal stratification will lead to an increase in the frequency, intensity and duration of cyanobacterial blooms across the globe (Huisman et al., 2018; Paerl and Huisman, 2008). It was therefore unexpected when neither the summer blooming genus *Microcystis* nor other cyanobacteria dominated the phytoplankton community in Lake Vombsjön, Sweden, during the unusually warm and dry summer of 2018, especially since cyanobacteria have dominated the phytoplankton community 11 out of 15 of the sampling years on record. Instead, the dinoflagellate *Naiadinitium polonicum*, for which there were no previous reports of blooms (Alström et al., 2017; Cronberg, 1996), dominated the phytoplankton

Table 2

Results from a model averaging (full average) for a multiple linear regression where environmental variables below were included. The criterion for a model to be included in model averaging was that $\Delta AIC < 2$. P-values in bold are considered significant (<0.05). NH_4 = ammonium, NO_3+NO_2 = nitrate + nitrite, Prec = 30-day cumulative precipitation, Q_{out} = 30-day average outflow, SRP = soluble reactive phosphorus, T_{1m} = surface water temperature, Z_{eu} = euphotic zone.

Response variable	Predictor variable	Coefficient	Standard error	P-value
<i>Naiadinitium polonicum</i>	NH_4	−0.18	0.11	0.11
	NO_3+NO_2	−0.003	0.06	0.95
	Prec	−0.11	0.11	0.35
	Q_{out}	−0.31	0.08	<0.001
	SRP	0.49	0.13	<0.001
	T_{1m}	0.48	0.12	<0.001
	Z_{eu}	−0.03	0.07	0.67
<i>Microcystis</i> spp.	NH_4	−0.03	0.05	0.60
	NO_3+NO_2	−0.23	0.07	<0.001
	Prec	0.02	0.06	0.66
	Q_{out}	0.06	0.06	0.37
	SRP	−0.02	0.05	0.77
	T_{1m}	0.03	0.06	0.61
	Z_{eu}	0.02	0.05	0.74

community from July 23rd onward. *N. polonicum* is a thecate dinoflagellate that has been identified in Europe, Israel, Japan, the Bahamas, Chile and Brazil (Björneras et al., 2020; Moestrup and Calado, 2018; Pollinger and Hickel, 1991). It is the only freshwater species known to produce ichthyotoxins that cause fish kills, though reported cases are rare (Hashimoto et al., 1968; Oshima et al., 1989; Roset et al., 2002), and thus few studies exist regarding *N. polonicum* and its autecology. Our overall findings, as discussed below, indicate that driving factors for this community shift included higher water temperatures, water column stability, and availability of phosphorus. We suggest that dinoflagellates, which share important functional traits with cyanobacteria, can outcompete cyanobacteria under certain climatic conditions, and that climate change, at least in temperate regions where water temperatures do not exceed 25°C , does not necessarily lead to an increase in cyanobacterial blooms.

4.1. Physical and chemical parameters

Weather conditions during 2018 had a strong impact on the physical conditions in Lake Vombsjön. Surface water temperatures increased but

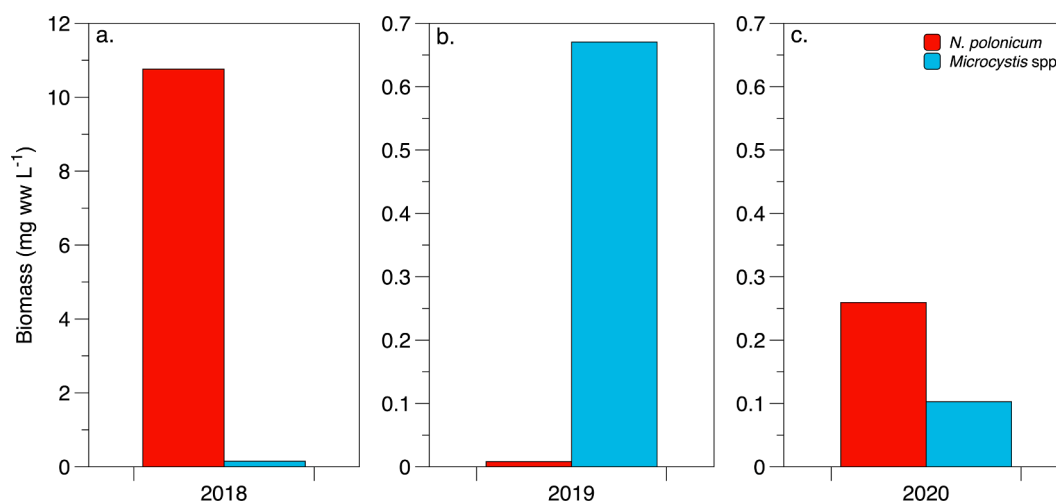


Fig. 8. Biomass assessed as area under the curve (AUC) of *Naiadinitium polonicum* and *Microcystis* spp. for 2018, 2019 and 2020 averaged over sampling season. Note that the y-axis in 2018 is different from 2019 to 2020.

never exceeded 25°C, which is often cited as a threshold temperature being particularly suitable for cyanobacteria blooms (Paerl and Huisman, 2008). In addition to warmer surface water temperatures, there was a stronger and more prolonged thermal stratification, and an extended period of hypoxia near the sediment surface when compared with subsequent years. In addition to the data collected here, a report on temperature profiles of Lake Vombsjön from 2014 to 2019 showed that apart from 2018 the lake rarely stratified (Holmström and Davidsson, 2021). Together, this information shows a drastic change in the physical conditions in Lake Vombsjön during 2018, particularly for the time period of May to July, which coincided with the recruitment and rapid increase of *N. polonicum* biomass. In terms of nutrient availability, it was expected that reduced precipitation would lead to reduced inflow, and therefore lower phosphorus and nitrogen concentrations in the lake. Total phosphorus and nitrogen concentrations were lower in 2018, but concentrations of soluble reactive phosphorus (SRP) were higher than in subsequent years. This was likely due to internal loading of SRP from the sediments as a consequence of low oxygen concentrations near the sediment surface (Mortimer, 1942). Thus, observed changes in the physical conditions of Lake Vombsjön during 2018 likely had an impact on chemical conditions as well.

4.2. Phytoplankton and zooplankton community composition

During our three-year study, cyanobacteria dominated the summer (August) phytoplankton community only in 2019 at which time *Microcystis* spp. had the largest biomass. *Microcystis* spp., while being present every year on record, was not as dominant in the historical data as previously thought, at least not since 1994. Instead, the historical phytoplankton community was often dominated by either *Woronichia naegeliana* or *Planktothrix agardhii*. Altogether, the success of *N. polonicum* during the heatwave of 2018 was in stark contrast to any other year on record, as was the success of green algae in 2020. The average surface water temperature in July 2020 (18.6 °C) was noticeably cooler than 2018 (22.3°C) or 2019 (20.2°C) likely benefitting green algae, which have their maximum growth rates at lower temperatures than do, for example, *Microcystis* (Johnk et al., 2008). *Naiadinitium polonicum* biomass increased rapidly in the fall of 2020, together with *Dolichospermum*, following an increase in temperature during the month of August. In fact, *Dolichospermum* dominated the counted community during the fall (September - October) of both 2019 and 2020, replacing fall dominance of *Planktothrix agardhii* in the historical record. Peak *Dolichospermum* biomass coincided with low DIN:SRP ratios in the lake, which is in line with studies showing that nitrogen-fixing cyanobacteria, like *Dolichospermum*, have a competitive advantage when nitrogen is limiting (Levine and Schindler, 1999; Vrede et al., 2009).

In addition to the phytoplankton community, we assessed the zooplankton community in order to understand whether grazing on *Microcystis* or other cyanobacteria may have contributed to the dominance of *N. polonicum* in 2018. Our study showed that copepods dominated the zooplankton community in 2018 and total biomass was higher than in subsequent years. However, our data does not suggest that grazing helped *N. polonicum* dominate in 2018, instead *N. polonicum* seemed to have a stronger influence on the zooplankton community than vice versa. Copepods, which are selective feeders, can graze on *Microcystis* but are considered to prefer eukaryotic phytoplankton to cyanobacteria as a food source (Ger et al., 2019). It is therefore likely that *N. polonicum*, which is an appropriate size (appx. 40 × 30 µm) for consumption by many copepod species (Hansen et al., 1994), was a good food source for copepods in Lake Vombsjön thus leading to higher biomass.

4.3. Dynamics of *Naiadinitium polonicum* and *Microcystis*

During our sampling, *Microcystis* and *N. polonicum* showed a strong presence/absence pattern, particularly in 2018 and 2019. The only

significant environmental predictor for *Microcystis* biomass in the multiple linear regression (MLR) with model averaging was nitrate plus nitrite, with which *Microcystis* was negatively correlated. In addition to nitrate plus nitrite, the principal component analysis showed *Microcystis* was negatively correlated with total nitrogen (TN) and TN:TP ratios and positively correlated with total phosphorus (TP); variables removed from the MLR because of collinearity. Many studies support a positive relationship between cyanobacteria biomass and both nitrogen and phosphorus, see for example Paerl et al. (2016). However, negative relationships between *Microcystis* biomass and nitrate concentrations have also been observed in lakes where blooms occur (Chaffin et al., 2011; Jacoby et al., 2000), and is likely the result of the uptake of nitrate by *Microcystis* as it grows. Compared with 2018 and 2020, nitrate plus nitrite concentrations were generally higher between May and July 2019, but phosphorus was at the same time limiting, so it is unlikely that nitrate plus nitrite in and of itself can explain the success of *Microcystis* in 2019.

The strongest environmental predictors for *N. polonicum* biomass were high surface water temperatures and SRP concentrations as well as low 30-day average outflow, all of which were strongly associated with the heatwave and drought that occurred in 2018. *N. polonicum* biomass was also highly associated with high relative water column stability (RWCS) (not included in the multiple linear regression), which is a proxy for thermal stratification, and hypoxia near the sediment surface.

Temperature is an extremely important factor affecting phytoplankton community assembly because of its direct effects on metabolic rates and therefore growth rates of phytoplankton. While little is known about *N. polonicum*, surface water temperatures during toxic blooms of *N. polonicum* in Japan 1963 (Hashimoto et al., 1968) and in Spain 1999 (Roset et al., 2002) were in the range 20–23°C. *N. polonicum* has also been found in warmer climates, including the Bahamas where temperatures were 23–26 °C (Björneras et al., 2020). Together these studies suggest that *N. polonicum* might have a proclivity for warmer temperatures thus benefitting them during the summer of 2018, though laboratory studies are needed to confirm this. Temperature is often cited as one of the main reasons cyanobacteria are expected to do better under future climate change scenarios, especially at temperatures above 25 °C where many eukaryotic species grow poorly (Paerl and Huisman, 2008). *Microcystis* growth rates increase with increasing temperature from a minimum temperature for growth of around 10–15°C to a temperature optima in the range of 27–32°C (Reynolds, 2006; Robarts and Zohary, 1987). In our study, we saw no relationship between higher water temperatures and the dominance of *Microcystis* or other cyanobacteria. Interestingly, other studies are also starting to challenge the paradigm that cyanobacteria do better at warmer temperatures, even at temperatures exceeding 25°C. For example, a survey of over 1000 lakes across the USA, spanning in surface water temperature from below 19°C to above 29°C, showed only a weak relationship between the relative abundance of cyanobacteria and temperature, and no relationship between the dominance of cyanobacteria and temperature (Verspagen et al., 2022). There was, however, a strong relationship between temperature and relative abundance of cyanobacteria on the genus level. For example, the relative abundances of the genera *Cylindrospermopsis*, *Planktolyngbya* and *Pseudanabaena* were all significantly higher in lakes with temperatures greater than 29°C (Verspagen et al., 2022). *Aphanizomenon*, on the other hand, decreased in lakes with temperatures above 24°C, while other genera, such as *Microcystis*, showed no differences in relative abundance between lake temperatures (Verspagen et al., 2022). Similarly, with the exception of polar climates, Bonilla et al. (2023) found no evident trends in cyanobacterial biomass with latitude or temperature based on data from 464 lakes along a 14,000 km north-south gradient. These recent papers, together with our findings, suggest that the relationship between cyanobacterial biomass and temperature in lakes may be more complicated than what is often presented in literature.

Rising surface water temperatures also have indirect effects on

phytoplankton, most notably in relation to the increased intensity and duration of thermal stratification of the water column, as seen in 2018 in Lake Vombsjön. One of the most important functional traits considered to benefit both cyanobacteria and dinoflagellates under stratified conditions are their ability to move vertically in the water column (Huisman et al., 2018; Smayda, 1997). While diurnal vertical migration patterns have been established for both dinoflagellates (Cullen, 1985; Smayda, 1997) and *Microcystis* (Ibelings et al., 1991; Kromkamp and Mur, 1984), nothing is currently known about vertical migration of *N. polonicum*. Furthermore, while studies have established *Microcystis* to be particularly adept at moving towards the surface after a mixing event in order to access light (Huisman et al., 2004), less is known regarding to what extent *Microcystis* can utilize their ability to sink in order to access nutrients from deeper waters. For example, Ibelings et al. (1991) found that diurnal patterns in *Microcystis* related to the day-night cycle was overruled by the influence of changes in mixing depth when relatively weak thermal stability was broken up by wind-induced turbulence. We suggest more information is needed in terms of the competitive ability for dinoflagellates and *Microcystis* to access nutrients from deeper waters under stratified conditions.

Soluble reactive phosphorus concentrations were strongly coupled to *N. polonicum* biomass in this study. Dinoflagellates are generally considered inferior when it comes to the uptake of phosphate compared with other groups of algae including cyanobacteria (Reynolds, 2006; Smayda, 1997). It is therefore possible that higher SRP concentrations in the water column benefitted *N. polonicum* which would otherwise be outcompeted by cyanobacteria. *Microcystis*, for example, are very effective at the uptake and storage of phosphate, even at low concentrations (Baldia et al., 2007; Harke et al., 2016; Kromkamp et al., 1989). Hypoxic conditions near the sediment surface could also have led to the internal loading of micronutrients, such as iron and manganese, from sediments (Mortimer, 1971). It is therefore also possible that nutrients not measured in this study benefitted *Naiadinitium* in 2018, similar to the 'unidentified growth factor' hypothesis from Zohary et al. (2012) for *Peridinium gatunense* blooms in Lake Kinneret, Israel.

In addition to positive relationships with temperature and soluble reactive phosphorus, *Naiadinitium* biomass was negatively correlated to one of the hydrological parameters included in our study: 30-day average outflow. Reduced outflow can increase water retention time and in turn the retention of phytoplankton biomass in lakes and reservoirs. Longer residence times have for example been associated with the relative success of the invasive, slow-growing, *Ceratium furcoides* in the Salto Grande Reservoir, Uruguay (Kruk et al., 2021). Lake Vombsjön, however, has a relatively long water residence time of about 200 days (Alström et al., 2017). It is therefore unlikely that reduced outflow affected biomass retention of *N. polonicum* directly. Instead, outflow was likely a co-variable with other drought factors such as reduced precipitation and increased temperature thus leading to a significant output from the multiple linear regression analysis.

5. Conclusion

In this study, *Naiadinitium polonicum* unexpectedly dominated during the extreme warm and dry summer of 2018, highlighting that competition between cyanobacteria and dinoflagellates can be an important factor when it comes to predicting cyanobacterial bloom dynamics under future climate change scenarios, particularly in temperate regions where temperatures often do not exceed 25°C. Key differences between 2018 and subsequent years were not only increased surface water temperature, but perhaps more importantly an extended period of high relative water column stability and hypoxia near the sediment surface, as well as higher SRP concentrations. *Naiadinitium polonicum*'s ecophysiology has not been studied in the laboratory yet, which makes it difficult to pinpoint which factors contributed to the shift in the summer phytoplankton community from being cyanobacteria- to *N. polonicum*-dominated. Furthermore, few studies have considered how freshwater

dinoflagellates, which share important functional traits with cyanobacteria, will be affected by climate change and extreme weather events. We suggest that more information is needed on the autecology and ecophysiology of *Naiadinitium polonicum*. Furthermore, the functional traits of cyanobacteria and dinoflagellates need to be explored further especially in relation to competition and niche partitioning in temperate lakes where temperatures can be optimal for both groups.

CRedit authorship contribution statement

Sandra Rabow: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Emma Johansson:** Writing – review & editing, Methodology, Investigation. **Per Carlsson:** Writing – review & editing, Supervision, Investigation. **Karin Rengefors:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hal.2024.102787.

Data availability

The data for this manuscript is available in the Dryad digital repository at <https://doi.org/10.5061/dryad.02v6wwq7s>.

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